



Multiple-Choice Questions (MCQs)

Sample set of 50 multiple-choice questions (MCQs) on the CPMAI Methodology,

- Each with **four options** and the **correct answer** marked.
- These cover fundamentals, phases, and data preparation concepts.

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CPMAI Fundamentals

1. CPMAI stands for:
A) Cognitive Project Management for AI ☒
B) Cognitive Process Modeling for AI
C) Continuous Project Management for AI
D) Cognitive Planning Model for AI
2. CPMAI integrates principles from:
A) Agile and Scrum
B) PMI and CRISP-DM ☒
C) Six Sigma and Lean
D) DevOps and Kanban
3. The CPMAI approach is primarily:
A) Linear
B) Iterative ☒
C) Random
D) Predictive
4. The main goal of CPMAI is to:
A) Automate all business processes
B) Align AI projects with business objectives ☒
C) Replace traditional project management
D) Focus only on data science
5. Which of the following is NOT a CPMAI phase?



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- A) Data Understanding
- B) Model Evaluation
- C) Model Deployment
- D) Data Visualization ☒

Phase I: Business Understanding

6. The first step in CPMAI is:
 - A) Data Preparation
 - B) Business Understanding ☒
 - C) Model Development
 - D) Model Evaluation
7. The key question in Business Understanding is:
 - A) Do we have enough data?
 - B) Do we need an AI approach? ☒
 - C) Which algorithm should we use?
 - D) How do we visualize results?
8. Business Understanding ensures:
 - A) Technical feasibility only
 - B) Business alignment and success criteria ☒
 - C) Data labeling accuracy
 - D) Model tuning
9. The output of this phase is:
 - A) Clean dataset
 - B) Business case and KPIs ☒
 - C) Trained model
 - D) Evaluation report
10. Stakeholder identification occurs in:



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- A) Model Development
- B) Business Understanding ☒
- C) Data Preparation
- D) Model Evaluation



Phase II: Data Understanding

11. The goal of Data Understanding is to:

- A) Train models
- B) Explore and assess data ☒
- C) Deploy models
- D) Define KPIs

12. Data profiling helps to:

- A) Identify data quality issues ☒
- B) Train algorithms
- C) Deploy models
- D) Create dashboards

13. Data bias detection occurs in:

- A) Data Understanding ☒
- B) Model Evaluation
- C) Model Development
- D) Business Understanding

14. The output of Data Understanding is:

- A) Data inventory and insights ☒
- B) Trained model
- C) Deployment plan
- D) Business case

15. Missing values and inconsistencies are identified in:



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- A) Data Preparation
- B) Data Understanding ☒
- C) Model Evaluation
- D) Model Operationalization

Phase III: Data Preparation

16.Data Preparation focuses on:

- A) Cleaning and structuring data ☒
- B) Deploying models
- C) Evaluating KPIs
- D) Creating dashboards

17.Data cleansing involves:

- A) Removing duplicates and fixing errors ☒
- B) Selecting algorithms
- C) Monitoring models
- D) Evaluating performance

18.Data labeling is used for:

- A) Unsupervised learning
- B) Supervised learning ☒
- C) Reinforcement learning
- D) Random sampling

19.Feature engineering means:

- A) Creating new variables to improve model accuracy ☒
- B) Removing irrelevant data
- C) Deploying models
- D) Evaluating KPIs

20.Data integration combines:



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A) Multiple data sources ☒

B) Algorithms

C) KPIs

D) Dashboards

21. The output of Data Preparation is:

A) Structured dataset ☒

B) Trained model

C) Evaluation report

D) Deployment plan

22. Poor data quality leads to:

A) Better model accuracy

B) Failed AI projects ☒

C) Faster deployment

D) Improved governance

23. Data transformation means:

A) Converting raw data into usable formats ☒

B) Labeling data

C) Deploying models

D) Evaluating KPIs

24. Data lineage ensures:

A) Traceability of data sources ☒

B) Faster training

C) Bias removal

D) Model tuning

25. Data preparation typically consumes:

A) 10–20% of project time

B) 80–90% of project time ☒



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C) 50% of project time

D) 30% of project time

Phase IV: Model Development

26. Model Development focuses on:

A) Building and training models ☒

B) Cleaning data

C) Evaluating KPIs

D) Deploying models

27. Algorithm selection occurs in:

A) Model Development ☒

B) Data Understanding

C) Model Evaluation

D) Business Understanding

28. Hyperparameter tuning improves:

A) Model performance ☒

B) Data quality

C) Business KPIs

D) Governance

29. Prototype creation happens in:

A) Model Development ☒

B) Model Evaluation

C) Data Preparation

D) Business Understanding

30. The output of Model Development is:

A) Candidate models ☒

B) Clean dataset

C) Business case



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D) Deployment plan

✓ Phase V: Model Evaluation

31. Model Evaluation ensures:

- A) Model meets business and technical criteria ✓
- B) Data is clean
- C) Algorithms are selected
- D) KPIs are defined

32. Accuracy and precision are tested in:

- A) Model Evaluation ✓
- B) Data Understanding
- C) Model Development
- D) Business Understanding

33. Bias detection occurs in:

- A) Model Evaluation ✓
- B) Data Preparation
- C) Model Operationalization
- D) Business Understanding

34. The output of Model Evaluation is:

- A) Evaluation report ✓
- B) Clean dataset
- C) Deployment plan
- D) Business case

35. Model validation compares results to:

- A) KPIs and metrics ✓
- B) Raw data
- C) Algorithms
- D) Dashboards



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Phase VI: Model Operationalization

36. Model Operationalization focuses on:

- A) Deploying and monitoring models ☒
- B) Cleaning data
- C) Evaluating KPIs
- D) Selecting algorithms

37. Governance and compliance are applied in:

- A) Model Operationalization ☒
- B) Data Preparation
- C) Model Evaluation
- D) Business Understanding

38. Continuous monitoring ensures:

- A) Model performance remains stable ☒
- B) Data is labeled
- C) KPIs are defined
- D) Algorithms are selected

39. Ethical AI practices are enforced in:

- A) Model Operationalization ☒
- B) Model Development
- C) Data Understanding
- D) Business Understanding

40. The output of Model Operationalization is:

- A) Operational AI solution ☒
- B) Clean dataset
- C) Evaluation report
- D) Business case



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Iterative Nature & Governance

41.CPMAI is iterative because:

- A) AI models need continuous improvement ☒
- B) Data never changes
- C) Business goals remain static
- D) Algorithms are fixed

42.Feedback loops connect:

- A) Model Operationalization to Data Understanding ☒
- B) Data Preparation to Model Evaluation
- C) Business Understanding to Model Development
- D) Model Evaluation to Governance

43.Governance ensures:

- A) Ethical and compliant AI usage ☒
- B) Faster deployment
- C) Data labeling
- D) Algorithm selection

44.Monitoring helps identify:

- A) Model drift ☒
- B) Data duplication
- C) KPI definition
- D) Algorithm bias

45.Iteration begins again when:

- A) New data or business changes occur ☒
- B) Model is perfect
- C) Deployment fails
- D) KPIs are unchanged



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General CPMAI Knowledge

46.CPMAI combines:

- A) Project management and data science ☒
- B) Marketing and sales
- C) Finance and operations
- D) HR and logistics

47.The methodology ensures:

- A) Business alignment and technical rigor ☒
- B) Faster coding
- C) Reduced data collection
- D) Elimination of human oversight

48.CRISP-DM contributes to CPMAI by:

- A) Providing a data-centric framework ☒
- B) Defining governance rules
- C) Automating deployment
- D) Simplifying KPIs

49.PMI contributes to CPMAI by:

- A) Providing project management structure ☒
- B) Automating data labeling
- C) Selecting algorithms
- D) Evaluating models

50.CPMAI success depends on:

- A) Collaboration between business and technical teams ☒
- B) Isolated data scientists
- C) Automated tools only
- D) Ignoring governance



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CPMAI Glossary

CPMAI : *Cognitive Project Management for AI* — a structured, iterative framework combining PMI project management and CRISP-DM data science methodology.

Business Understanding : The first phase, focused on defining objectives, success criteria, and determining whether AI is the right approach.

Data Understanding : The phase where data sources are explored, profiled, and assessed for quality, completeness, and relevance.

Data Preparation : The process of cleaning, labeling, transforming, and integrating data into a usable format for modeling.

Model Development : Building, training, and tuning AI models using prepared datasets and selected algorithms.

Model Evaluation : Validating models against technical metrics and business KPIs to ensure readiness for deployment.

Model Operationalization : Deploying AI models into production, monitoring performance, and applying governance and compliance standards.

Data Cleansing : Removing duplicates, fixing errors, and handling missing values to improve dataset quality.

Data Labeling : Annotating data for supervised learning tasks to guide model training.

Feature Engineering : Creating new variables or features that enhance model accuracy and performance.

Data Integration : Combining multiple data sources into a unified dataset for analysis and modeling.

Iterative Cycle : The repeating loop of phases that ensures AI solutions evolve with new data, feedback, and business changes.

Feasibility Assessment : Evaluating vendor, culture, process, data, market, implementation, and business feasibility before committing to AI projects.

Governance : Oversight mechanisms ensuring ethical, compliant, and responsible AI deployment.



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<u>Term / Phase</u>	<u>Definition</u>	<u>Key Activities</u>	<u>Outputs</u>
CPMAI	Cognitive Project Management for AI — a framework combining PMI project management and CRISP-DM data science methodology.	Align AI projects with business goals, iterative cycle, governance.	Structured AI project roadmap.
Business Understanding	Defines objectives, success criteria, and AI feasibility.	Identify stakeholders, define KPIs, assess AI need.	Business case, project charter, success metrics.
Data Understanding	Explores and assesses available data for quality and relevance.	Collect datasets, profile data, detect bias, identify gaps.	Data inventory, initial insights, feasibility report.
Data Preparation	Cleans, labels, transforms, and integrates data for modeling.	Data cleansing, Data labeling, Feature engineering, Data integration.	High-quality structured dataset.
Model Development	Builds and trains AI models using prepared data.	Select algorithms, train models, tune hyperparameters, prototype creation.	Candidate models with performance metrics.
Model Evaluation	Validates models against technical metrics and business KPIs.	Test accuracy, precision, recall, bias detection, KPI comparison.	Evaluation report, deployment decision.
Model Operationalization	Deploys AI models into production with monitoring and governance.	Deploy models, monitor performance, apply governance, ensure compliance.	Operational AI solution with monitoring plan.
Iterative Cycle	Continuous loop of phases ensuring AI evolves with new data and business changes.	Feedback loops, retraining, refinement.	Updated models and processes.
Feasibility Assessment	Evaluates vendor, culture, process, data, market, implementation, and business feasibility.	Conduct AI Go/No-Go decision checks.	Feasibility report, project approval/rejection .
Governance	Oversight ensuring ethical, compliant, and responsible AI deployment.	Apply compliance standards, monitor ethics, enforce accountability.	Governance framework, compliance reports.